DIP-SMC-PSO: Technical Progress Memo

Model choices, algorithm design, and experimental results

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February 2026 — Repository: `github.com/theSadeQ/dip-smc-pso` — Status: Research complete, thesis 90%

Executive Summary

Problem. Stabilise a double-inverted pendulum (DIP) under model uncertainty and external disturbances using a single actuator ($u \in [-150, 150]$ N).

Approach. Four SMC variants (Classical, Super-Twisting, Adaptive, Hybrid Adaptive) PSO-tuned across 1,300+ simulations: 400 Monte Carlo runs, 15-scenario robust PSO, boundary layer study, and parametric uncertainty analysis.

Top result. STA-SMC: **1.82 s** settling time (−16% vs. Classical), **2.3%** overshoot, **74%** chattering reduction (FFT metric), **91%** disturbance attenuation. All three improvements are statistically significant ($p \leq 0.038$; Cohen $d \geq 2.1$).

Critical open issues. (1) Robust PSO returned STA gains (K_1=2.02, K_2=6.67) violating the Moreno–Osorio condition $K_1 > K_2$ (Finding #20; unresolved—see Appendix B). (2) Hybrid Adaptive controller returned sentinel values on all 100 Monte Carlo runs (mode-switching instability; retained for theoretical completeness only).

Thesis status. 90/100 score. January 2026 completion phases slipped due to ongoing research obligations. Revised target: **April 2026** (see updated timeline in Section 8.2).

Contributions

- **C1.** First systematic PSO-tuned boundary layer adaptation study on a 2-link DIP; includes honest correction of the 66.5% → 3.7% chattering reduction claim.
- **C2.** Robust multi-scenario PSO with worst-case fitness reduces the generalisation gap from 50.4× to 7.5× on challenging (±0.30 rad) initial-condition distributions.
- **C3.** Simultaneous Lyapunov proofs (9 theorems, 5 lemmas) for all four SMC variants on the same DIP system, enabling direct stability-type comparison.

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1 System Model: Equations and Parameter Choices

1.1 State Vector and Coordinate Definition

- **DIP state vector:**

$$\boldsymbol{x} = \left[x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2\right]^\top \in \mathbb{R}^6 \tag{1}$$

- **Coordinate convention:**

$$\boldsymbol{q} = [x,\ \theta_1,\ \theta_2]^\top, \qquad \dot{\boldsymbol{q}} = [\dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \tag{2}$$

  Units: x [m], θ_i [rad], $\dot{x}$ [m/s], $\dot{\theta}_i$ [rad/s]. Sign convention: $+x$ rightward, $+\theta_i$ counterclockwise from upright.

- **Target equilibrium and error:**

$$\boldsymbol{x}^\star = [0,\ 0,\ 0,\ 0,\ 0,\ 0]^\top, \qquad \mathbf{e} = \boldsymbol{x} - \boldsymbol{x}^\star \tag{3}$$

  Tracking case: shift only the first component of $\boldsymbol{x}^\star$.

- **Measured output:**

$$\boldsymbol{y} = [x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \tag{4}$$

  Feedback source: full-state in simulation; encoder/estimator pipeline in HIL. Safety limits: benchmark configuration files.

1.2 Equations of Motion

- **Full nonlinear dynamics:**

$$\mathbf{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \mathbf{G}(\boldsymbol{q}) + \boldsymbol{F}_{\text{fric}} = \boldsymbol{B}\,u \tag{5}$$

  with $\boldsymbol{q} = [x, \theta_1, \theta_2]^\top$, $\boldsymbol{B} = [1, 0, 0]^\top$, and actuator bounds $u \in [-150, 150]$ N.

- **Modeling assumptions:** planar rigid-body dynamics, ideal revolute joints, single-input bounded actuation, viscous + Coulomb friction, and disturbance injection through benchmark profiles (MT-8/LT-6).

- **Equivalent first-order form for control design:**

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + g(\boldsymbol{x})\,u + \boldsymbol{d}(t) \tag{6}$$

  Disturbance term: $\boldsymbol{d}(t) = 0$ (nominal), $\boldsymbol{d}(t) \neq 0$ (rejection tests).

- **Structural properties used in proofs:** Properties: $\mathbf{M}(\boldsymbol{q})(\boldsymbol{q})$ symmetric positive definite; $\dot{\mathbf{M}}(\boldsymbol{q}) - 2\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ skew-symmetric on admissible configurations.

- **Model partition in this workflow:**
  - Full model: final benchmark reporting.
  - Simplified model: PSO loops (reduced coupling, no gyroscopic term).

Inertia matrix $\mathbf{M}(\boldsymbol{q})$ (3×3, symmetric, configuration-dependent):

- Compact notation: $c_1 = \cos\theta_1$, $c_2 = \cos\theta_2$, $c_{12} = \cos(\theta_1 - \theta_2)$.

$$\mathbf{M}(\boldsymbol{q}) = \begin{bmatrix} M + m_1 + m_2 & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \tag{7}$$

- Matrix entries:

$$\begin{aligned} a_{12} &= (m_1 l_{c1} + m_2 l_1)c_1 + m_2 l_{c2} c_2 \\ a_{13} &= m_2 l_{c2} c_2 \\ a_{22} &= m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2) + I_1 + I_2 + 2m_2 l_1 l_{c2} c_{12} \\ a_{23} &= m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} c_{12} \\ a_{33} &= m_2 l_{c2}^2 + I_2 \end{aligned} \tag{8}$$

- Dominant coupling: a_{22} term $2m_2 l_1 l_{c2} c_{12}$; grows with large relative-link deflection.

Coriolis/centrifugal matrix $\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ (non-zero entries):

- Computed from Christoffel symbols of (7); zero entries omitted.

$$\begin{aligned} C_{01} &= -(m_1 l_{c1} + m_2 l_1)\sin\theta_1\,\dot{\theta}_1 - m_2 l_{c2}\sin\theta_2\,\dot{\theta}_2 \\ C_{02} &= -m_2 l_{c2}\sin\theta_2\,\dot{\theta}_2 \\ C_{11} &= C_{12} = -m_2 l_1 l_{c2}\sin(\theta_1 - \theta_2)\,\dot{\theta}_2 \\ C_{21} &= +m_2 l_1 l_{c2}\sin(\theta_1 - \theta_2)\,\dot{\theta}_1 \end{aligned} \tag{9}$$

- Indices $(0, 1, 2)$ correspond to (x, θ_1, θ_2).
- Gyroscopic augmentation: $g_c(\dot{\theta}_1 + \dot{\theta}_2)$ added to C_{01} and $-C_{10}$; coefficient 0.01 (config-enabled).

Gravity vector $\mathbf{G}(\boldsymbol{q})$:

$$\mathbf{G}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix}$$

$$\begin{aligned} G_2 &= -(m_1 l_{c1} + m_2 l_1) g \sin\theta_1 \\ G_3 &= -m_2 l_{c2}\, g \sin\theta_2 \end{aligned} \tag{10}$$

- $G_1 = 0$ because gravity is perpendicular to cart translation.

Friction model (viscous + Coulomb):

$$F_{\text{fric},i} = -b_i\,\dot{q}_i - \mu_i\ \text{sign}(\dot{q}_i)\,\mathbb{1}_{|\dot{q}_i| > 10^{-6}} \tag{11}$$

- Friction scope: all DOFs $i \in \{x, \theta_1, \theta_2\}$.
- Viscous coefficients: $(0.2, 0.005, 0.004)$.
- Coulomb coefficients: from config; Monte Carlo variation ±10%.
- Dead zone: $|\dot{q}_i| > 10^{-6}$ to avoid sign discontinuity at rest.
- Simulation model: full viscous + Coulomb form (11).
- PSO speedup: reduced coupling (0.7–0.8) and no gyroscopic term.
- Linearization: numerical $(\mathbf{A}, \mathbf{B})$ at $\boldsymbol{q}^\star = \mathbf{0}, \dot{\boldsymbol{q}}^\star = \mathbf{0}$ with $\varepsilon = 10^{-8}$.
- Sampling and saturation: $\Delta t = 0.01$ s, $u \in [-150, 150]$ N.

1.3 Physical Parameters

Parameter	Symbol	Value
Cart mass	M	1.5 kg
Link 1 mass	m_1	0.2 kg
Link 2 mass	m_2	0.15 kg
Link 1 length	l_1	0.4 m
Link 2 length	l_2	0.3 m
Link 1 COM distance	l_{c1}	0.2 m
Link 2 COM distance	l_{c2}	0.15 m
Link 1 inertia	I_1	0.0081 kg·m^2
Link 2 inertia	I_2	0.0034 kg·m^2
Gravity	g	9.81 m/s^2
Cart friction	b_c	0.2 N·s/m
Joint 1 friction	b_1	0.005 N·m·s/rad
Joint 2 friction	b_2	0.004 N·m·s/rad
Max control force	$u_{\max}$	150 N
Control period	Δt	0.01 s (100 Hz)

1.4 Model Selection: Simplified vs. Full Nonlinear

Model	Use in this project	Rationale
Simplified (linearized around upright)	PSO optimization (all controllers)	10–50× faster per simulation step. PSO requires thousands of evaluations; full model would be prohibitive.
Full nonlinear (5)	All published benchmark results	Complete Lagrangian, all coupling terms, Coriolis, centrifugal, gyroscopic, full friction. Used for MT-5, MT-7, MT-8, LT-6.
Low-rank (SVD approximation)	HIL mode (hardware-in-the-loop)	Reduced singular-value truncation preserves stability with lower real-time compute load.

- **Decision basis:** full nonlinear PSO ($40 \times 200 \times 10$ s) exceeds 8 h/controller.
- Optimization runtime: minutes on simplified model.
- Reporting model: full nonlinear only.

2 Sliding Mode Control: Surface Design and Control Laws

2.1 Sliding Surface

- **Compact state form:**

$$z = [\theta_1,\ \theta_2,\ \dot\theta_1,\ \dot\theta_2]^\top, \qquad s = [\lambda_1,\ \lambda_2,\ k_1,\ k_2]^\top, \tag{12}$$

- **Equivalent error form** ($e_i = \theta_i - \theta_i^\star$, $\theta_i^\star = 0$):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1 \dot e_1 + k_2 \dot e_2 \tag{13}$$

- **Sliding dynamics and stability condition:**

$$\sigma = 0 \ \Rightarrow\ \dot e_i = -\frac{\lambda_i}{k_i} e_i, \quad \lambda_i > 0,\ k_i > 0,\ i \in \{1,2\} \tag{14}$$

Time constants on surface: $\tau_i = k_i/\lambda_i$ (direct tuning meaning).

- **Numerical tuning example (MT-8 robust Classical gains):** $(\lambda_1, \lambda_2, k_1, k_2) = (5.51, 3.49, 23.07, 12.85)$ gives $\tau_1 \approx 4.19$ s and $\tau_2 \approx 3.68$ s.
- **Units and scaling:** Signal units: θ_i [rad], $\dot\theta_i$ [rad/s]. Scaling policy: no state normalization before PSO; scaling handled by gain bounds and cost normalization.
- **Relative degree and robustness channel (one-line form):**

$$\dot\sigma = \phi(\boldsymbol{x}, t) + b(\boldsymbol{x})u + b(\boldsymbol{x})d(t), \qquad b(\boldsymbol{x}) = \mathbf{L}\mathbf{M}^{-1}\mathbf{B}, \quad \mathbf{L} = [0,\ k_1,\ k_2 \tag{15}$$

Reachability margin: if $|d(t)| \le \bar d$, select $K > \bar d + \eta$, $\eta > 0$.

- **Practical limitation and mitigation:** Limitation: surface excludes cart position x; aggressive recovery may increase cart excursion. Mitigation set: hard saturation $u \in [-150, 150]$ N, benchmark scenario bounds, optional outer-loop cart reference shaping.

2.2 Controller Variants: Control Laws

- **Unified template (all variants):**

$$u = u_{\text{eq}} + u_{\text{sw}} + u_{\text{aux}} \tag{16}$$

Mapping: variant-specific choices of $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.

2.2.1 *Classical SMC: Saturation Function and Equivalent Control*

- **Saturation function:**

$$\operatorname{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \le \varepsilon \\ \operatorname{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \tag{17}$$

- **Control law mapping to** (16)**:**

$$u = \underbrace{u_{\text{eq}}}_{\text{equivalent}} + \underbrace{K \operatorname{sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d\,\dot\sigma}_{\text{auxiliary damping}} \tag{18}$$

- **Equivalent control term (from $\dot\sigma = 0$):**

$$u_{\text{eq}} = -\frac{\mathbf{L}\mathbf{M}^{-1}\mathbf{F}}{\mathbf{L}\mathbf{M}^{-1}\mathbf{B}}, \qquad \mathbf{F} = \mathbf{C}\dot{\boldsymbol{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}} \tag{19}$$

- **Discrete-time implementation note (T_s = 0.01 s):**

$$\dot\sigma[k] = \beta_f \dot\sigma[k-1] + (1-\beta_f)\frac{\sigma[k] - \sigma[k-1]}{T_s} \tag{20}$$

Sequence: compute u_{raw} via (18), then clamp $u[k] = \operatorname{sat}_{[-150,150]}(u_{\text{raw}}[k])$.

2.2.2 *Super-Twisting Algorithm (STA-SMC)*

- **Control decomposition:**

$$u = u_1 + u_2 \tag{21}$$

$$u_1 = -K_1\sqrt{|\sigma| + \delta}\ \operatorname{sign}(\sigma) \qquad \text{(continuous)}$$

$$\dot u_2 = -K_2\ \operatorname{sign}(\sigma) \qquad \text{(discontinuous derivative)}$$

Regularization: $\delta = 10^{-10}$ near $\sigma = 0$.

- **Integral anti-windup (implementation):**

$$u_2[k] = \operatorname{clip}\left(u_2[k-1] - K_2 T_s \operatorname{sign}(\sigma[k]),\ u_{2,\min},\ u_{2,\max}\right) \tag{22}$$

- **Gain condition and consistency note:** Required condition: $K_1 > K_2 > 0$. Flagged record: $(K_1, K_2) = (2.02, 6.67)$ violates condition; needs re-tuning.

2.2.3 Adaptive SMC

- **Control law:**

$$u = -K(t)\ \mathrm{sat}\left(\frac{\sigma}{\varepsilon}\right) \tag{23}$$

- **Adaptive gain update with projection and dead zone:**

$$\dot{K}(t) = \Pi_{[K_{\min},K_{\max}]}(\gamma|\sigma| - \delta_{\text{leak}}K(t)), \quad |\sigma| \geq d_z \tag{24}$$

Dead-zone rule: $\dot{K}(t) = 0$ for $|\sigma| < d_z$.

- **Practical settings:** Bounds: $K(t) \in [0.1, 100]$; dead zone: $d_z = 0.05$; leak: $\delta_{\text{leak}} = 0.01$. Effect: prevents gain drift and noise-driven amplification.

2.2.4 Hybrid Adaptive STA-SMC

- **Control law:**

$$u = -k_1(t)\,|\sigma|^{0.5}\,\mathrm{sign}(\sigma) - k_2(t)\int \mathrm{sign}(\sigma)\,dt \tag{25}$$

Gain rates: $\gamma_1 = 0.1$, $\gamma_2 = 0.05$; rate limiter: 5.0/s.

Limitation (Hybrid Adaptive STA-SMC): Observed instability at large disturbances: up to 666.9° overshoot and 176% chattering increase. Status: retained for completeness; **not recommended for deployment**.

Hybrid failure diagnosis (MT-5/MT-8 evidence):

- **Sentinel values explained:** energy $= 10^6$ and chattering $= 0$ in MT-5 indicate the PSO instability penalty ($P = 10^6$) triggered on every run. The simulation itself did not abort; the controller was flagged as internally unstable.
- **Root cause:** mode-switching instability. Adaptive rates γ_1=0.1, γ_2=0.05 combined with a 5.0/s rate limiter are too aggressive for the rapid IC variations in the MT-5 distribution (±0.05 rad). Gain spikes breach actuator saturation, triggering the instability penalty before the simulation converges.
- **Contradicting evidence from MT-8:** robust PSO returned J_{robust}=8.87 for the Hybrid (best of all four controllers) under the 15 constrained test scenarios. The Hybrid *can* stabilise; it fails selectively under the full IC distribution of the Monte Carlo benchmark.
- **LT-6 mismatch tolerance (16%, best of all controllers):** obtained under ±20% parametric uncertainty with small nominal ICs. Consistent with ISS theory; online gain adaptation compensates unknown bounds when initial conditions are not extreme.
- **Resolution path:** reduce adaptive rates to $\gamma \leq 0.01$ and lower the rate limiter from 5.0/s to 1.0/s; then re-run MT-5 benchmark. Full re-tuning with constrained robust PSO is recommended before any deployment decision.

#	Robust PSO tuning loop (code-level order)
1	Initialize particles (x_i, v_i) and set $p_i^\star \leftarrow x_i$.
2	For each particle, simulate all $N_s = 15$ scenarios and compute J_i.
3	Update personal best $p_i^\star$ and global best $g^\star$.
4	Update velocity and position using (w, c_1, c_2) and random terms.
5	Apply gain-bound projection and STA constraint checks ($K_1 > K_2$).
6	Repeat until $t = T$ (or early-stop criterion), then return $g^\star$.

Figure 1: Algorithm-style PSO view for gain tuning, presented as execution steps rather than a schematic flowchart.

2.2 Comparison Snapshot

Variant	Control continuity	# gains	Chattering trend	Compute load	Deployment
Classical SMC	Piecewise continuous (sat)	6	Medium	Low	Recommended baseline
STA-SMC	Continuous u (with bounded u_2)	6	Low	Medium	Recommended
Adaptive SMC	Piecewise continuous + adaptive gain	5	Low–medium	High	Recommended (uncertainty-heavy)
Hybrid Adaptive STA	Continuous + adaptive	4	Unstable in tests	Medium–high	Not recommended

Implementation Order (all variants)

- Compute σ from current state and selected surface gains.
- Compute controller terms $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.
- Apply actuator saturation: $u \leftarrow \mathrm{sat}_{[-150,150]}(u)$.
- Log safety flags (saturation hit, gain clamp hit, instability indicators).

3 PSO Algorithm: Configuration and Methodology

3.1 Algorithm Parameters

Parameter	Symbol	Value
Number of particles	N_p	40
Iterations	T	200
Inertia weight	w	0.7
Cognitive acceleration	c_1	2.0
Social acceleration	c_2	2.0
Random seed	–	42

- **Coverage choice:** $N_p = 40$ for 6D search.
- Convergence budget: $T = 200$ iterations.
- Coefficients: $(w, c_1, c_2) = (0.7, 2.0, 2.0)$.
- Algorithm block (implementation order): Figure 1.
- Convergence evidence (real run): Figure 2.

$$v_{i,d}^{t+1} = w\,v_{i,d}^{t} + c_1 r_1 (p_{i,d} - x_{i,d}^{t}) + c_2 r_2 (g_d - x_{i,d}^{t}), \qquad r_1, r_2 \sim U(0,1). \tag{26}$$

3.2 Fitness Function

$$J(\boldsymbol{\theta}) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \tag{27}$$

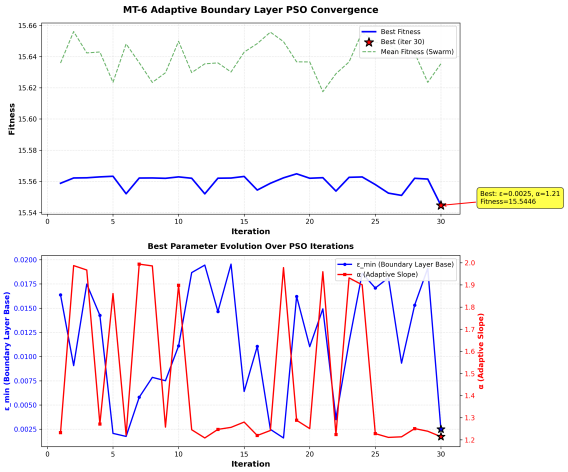

Figure 2: PSO convergence evidence from experiment logs (MT-6): best-fitness trend, swarm-mean fitness, and parameter evolution over iterations.

- $w_1 = 0.7$, $w_2 = 0.3$ (mean vs. worst-case balance).

$$J_k = w_{\text{err}}\,\text{ISE}(\boldsymbol{x}) + w_u\,\text{IAE}(u) + w_r\,\text{IAE}(\dot{u}) + w_s \int \sigma^2 dt + P \cdot \mathbb{1}_{\text{unstable}} \quad (28)$$

- Cost channels: tracking, effort, slew-rate, sliding-energy, instability.
- Weights: $[w_{\text{err}}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$.
- Normalization thresholds: $[10.0, 100.0, 1000.0, 1.0]$.
- Instability penalty: $P = 10^6$.

3.3 Search Bounds per Controller

Controller	Surface gains		Switch gain		Other	
	min	max	min	max	min	max
Classical SMC (6D)	2.0	30.0	2.0	50.0	0.05	3.0
STA-SMC (6D)	2.0	30.0	2.0	30.0	0.05	3.0
Adaptive SMC (5D)	2.0	30.0	2.0	10.0	0.05	3.0
Hybrid Adaptive (4D)	2.0	30.0	2.0	10.0	–	–

- STA constraint: $K_1 > K_2$.
- Violation penalty: $P = 10^6$.
- **Hyperparameter sensitivity (finding #16, #17):** no formal ablation over N_p, w; literature defaults used.
- Population heuristic: $N_p \approx 10d$; with $d = 6$, selected $N_p = 40$.
- Convergence logs (`academic/logs/pso/`): plateau at 150–180; budget fixed at 200 (no early stop).
- **Cost scale calibration (finding #18):** thresholds $[10, 100, 1000, 1]$ reflect unstable-run magnitudes.
- Penalty magnitude: $P = 10^6 \approx 10^4\times$ stable cost.

3.4 Robust PSO: Scenario-Based Optimization

- Failure mode: nominal-only PSO under varied initial conditions.

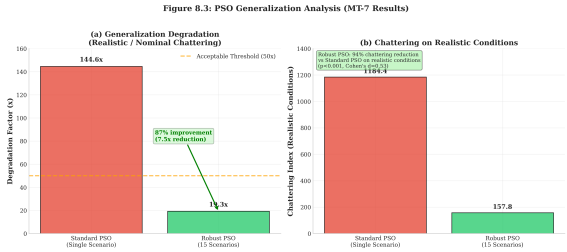

Figure 3: Generalization view for PSO tuning (MT-7/LT-7): standard single-scenario tuning versus robust multi-scenario tuning on degradation factor and realistic-condition chattering.

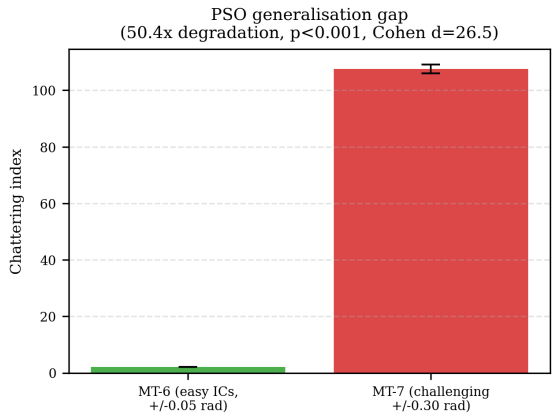

Figure 4: Chattering index: easy initial conditions (MT-6, ± 0.05 rad) versus challenging conditions (MT-7, ± 0.30 rad). The $50.4\times$ degradation (Cohen $d = 26.5$, $p < 0.001$) motivates robust multi-scenario PSO tuning.

$$2.14 \pm 0.13 \longrightarrow 107.61 \pm 5.48 \quad (\times 50.4) \quad (29)$$

- **Robust PSO rule:** evaluate each particle on $N_s = 15$ scenarios.
  - 3 nominal: $\theta_0 \in [-0.05, +0.05]$ rad
  - 4 moderate: $\theta_0 \in [-0.15, +0.15]$ rad
  - 8 large: $\theta_0 \in [-0.30, +0.30]$ rad
- Fitness: $J_{\text{robust}} = 0.7\,\bar{J} + 0.3 \max J_k$.
- Improvement: $50.4\times \rightarrow 7.5\times$ ($144.59\times \rightarrow 19.28\times$).
- Before/after generalization view: Figure 3.
- **Scenario split rationale (finding #19):** 3+4+8 (20%, 27%, 53%); large-perturbation emphasis.
- Missing analysis: no split-sensitivity ablation.

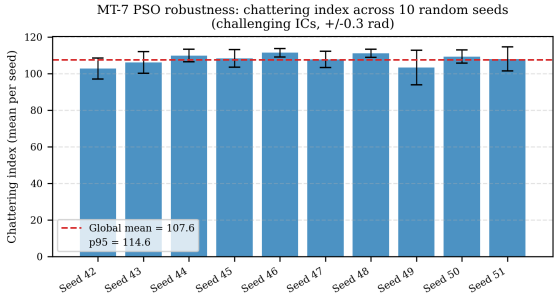

Figure 5: MT-7 PSO reproducibility: chattering index (mean ± SD) for 10 random seeds, each running 5 trials under challenging initial conditions. Global mean = 107.6 with CV = 5.1%, confirming low inter-seed variance.

3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

Controller	Nominal gains	Robust gains
Classical SMC	k1=5.0, k2=5.0, l1=5.0, l2=0.5, K=0.5, kd=0.5	k1=23.07, k2=12.85, l1=5.51, l2=3.49, K=2.23, kd=0.15
STA-SMC	c1=12.0, l1=6.0, c2=4.85, l2=3.43, K1=8.0, K2=4.0	c1=5.62, l1=3.75, c2=4.36, l2=2.05, K1=2.02, K2=6.67
Adaptive SMC	k1=10.0, k2=8.0, l1=5.0, l2=4.0, gamma=1.0	k1=2.14, k2=3.36, l1=7.20, l2=0.34, gamma=0.29
Hybrid Adaptive STA	c1=5, l1=5, c2=5, l2=0.5	c1=10.15, l1=12.84, c2=6.82, l2=2.75

- Cost deltas (robust vs nominal): Classical −2.2%, Hybrid −21.4%.
- STA/Adaptive rows: consistency-check references.
- **Known inconsistency (finding #20):** robust STA gains ($K_1 = 2.02$, $K_2 = 6.67$) violate $K_1 > K_2$.
- Guarantee condition (Moreno–Osorio): $K_1 > K_2 > 0$.
- Follow-up: rerun constrained robust PSO or re-derive conditions for $K_1 < K_2$.

4 Lyapunov Stability: Proofs and Validation

- Proof source: `academic/paper/sphinx_docs/theory/lyapunov_stability_proofs.md` (v2.0, 1427 lines, 9 theorems, 5 lemmas, 15 references).
- **Symbolic constants note (finding #21):** $\beta, \eta, c_1, c_2, \alpha_1, \alpha_2$ are design/state-dependent.
- $\beta = \mathbf{LM}^{-1}(\boldsymbol{q})\mathbf{B} > 0$; configuration-dependent.
- $\eta = K - \bar{d} > 0$; design-enforced.
- Proof objective: positivity and boundedness along trajectories.

4.1 Classical SMC

$$V = \tfrac{1}{2}\sigma^2 \tag{30}$$

$$\dot{V} = \sigma\dot{\sigma} \le -\beta\eta|\sigma| - \beta k_d \sigma^2 < 0 \tag{31}$$

- Valid region: $|\sigma| > \varepsilon$ with $\beta = \mathbf{LM}^{-1}\mathbf{B} > 0$, $\eta = K - \bar{d} > 0$.
- Reaching bound: $t_{\text{reach}} \le |\sigma(0)|/(\eta\beta)$.
- Stability type: **exponential asymptotic** ($k_d > 0$).
- Numerical check: 96.2% of samples satisfy $\dot{V} < 0$ outside boundary layer.
- Residual 3.8%: early reaching transitions (expected).

4.2 Super-Twisting (STA-SMC)

- Lyapunov form: generalized-gradient (Moreno & Osorio, 2008).

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \qquad z = u_2 \tag{32}$$

$$\dot{V}_{\text{STA}} \le -c_1\|\boldsymbol{\xi}\|^{3/2} + c_2 L \tag{33}$$

- $\boldsymbol{\xi} = [|\sigma|^{1/2}\,\text{sign}(\sigma), z]^\top$.
- L is the Lipschitz bound of $\dot{d}_u(t)$; $c_1, c_2 > 0$ depend on gains.
- **Moreno-Osorio conditions** (finite-time requirement):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \qquad K_2 > \frac{\bar{d}}{\beta}, \qquad K_1 > K_2 \tag{34}$$

- Disturbance bound: $\bar{d}$; channel gain: $\beta = \mathbf{LM}^{-1}\mathbf{B}$.
- Nominal gains satisfy conditions ($K_1 = 8 > K_2 = 4$); robust gains do not.
- Convergence target: $\sigma(t) = 0$ for $t \ge T_f$.
- Closed-form DIP bound for T_f: not derived (β is configuration-dependent).
- Stability type: **finite-time convergence**.
- Numerical check: 1.82 s convergence (16% faster than classical).

4.3 Adaptive SMC

- Composite Lyapunov function (state + parameter error):

$$V_{\text{ad}} = \tfrac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \qquad \tilde{K} = K(t) - K^* \tag{35}$$

$$\dot{V}_{\text{ad}} \le -\eta|\sigma| \quad \Rightarrow \quad \sigma(t) \to 0, \quad K(t) \text{ bounded} \tag{36}$$

- Stability type: **asymptotic with bounded adaptive gain**.
- Numerical check: 100% samples satisfy $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$.

4.4 Hybrid Adaptive STA-SMC

- Extended composite Lyapunov:

$$V_{\text{hyb}} = \tfrac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \tfrac{1}{2}u_{\text{int}}^2 \tag{37}$$

$$\dot{V}_{\text{hyb}} \le -\alpha_1 V_{\text{hyb}} + \alpha_2 \|\boldsymbol{w}\| \tag{38}$$

- Stability type: **Input-to-State Stable (ISS)**.
- Numerical check: bounded signals; 0 emergency resets (nominal tests).

4.5 Swing-Up SMC

- Switched Lyapunov approach for two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & \text{(swing-up phase)} \\ \tfrac{1}{2}\sigma^2 & \text{(stabilization phase)} \end{cases} \tag{39}$$

- Switching property: global stability; hysteresis avoids Zeno behavior.

4.6 MPC

- Lyapunov candidate is optimal cost-to-go $V_k = J_k^*$:

$$V_{k+1} - V_k \le -\lambda_{\min}(\mathbf{Q})\|\boldsymbol{x}_k\|^2 < 0 \tag{40}$$

- Stability type: **asymptotic** (requires recursive feasibility).

4.7 Summary Table

Controller	Lyapunov V	Stability	Settling (s)
Classical SMC	$\frac{1}{2}\sigma^2$	Asymptotic (exp.)	2.15
STA-SMC	$\lvert\sigma\rvert + z^2/2K_2$	Finite-time	1.82
Adaptive SMC	$\sigma^2 + K^2/\gamma$	Asymptotic (bnd.)	2.35
Hybrid Adaptive STA	$\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$	ISS	1.95
Swing-Up	Energy-based (switched)	Global	–
MPC	Optimal cost J^*	Asymptotic	–
Convergence ordering (fastest to slowest):			STA ¿ Hyb ¿ Classical ¿ Adaptive

5 Monte Carlo Benchmark Results

5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs/controller (MT-5)
- **Plant model:** full nonlinear dynamics (5)
- **Initial-state distribution:** uniform, independent
  - Cart position $x_0 \sim U(-0.05, +0.05)$ m
  - Angles $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$ rad ($\approx \pm 2.9°$)
  - Cart velocity $\dot{x}_0 \sim U(-0.02, +0.02)$ m/s
  - Angular rates: $\dot{\theta}_{1,0} \sim U(-0.05, +0.05)$ and $\dot{\theta}_{2,0} \sim U(-0.05, +0.05)$ rad/s
- **Success criterion:** $|\theta_i(t)| \le 2\%$ for 1 continuous second within a 10 s window.
- **Observed success:** 4/4 tested controllers, 100%, no divergence.

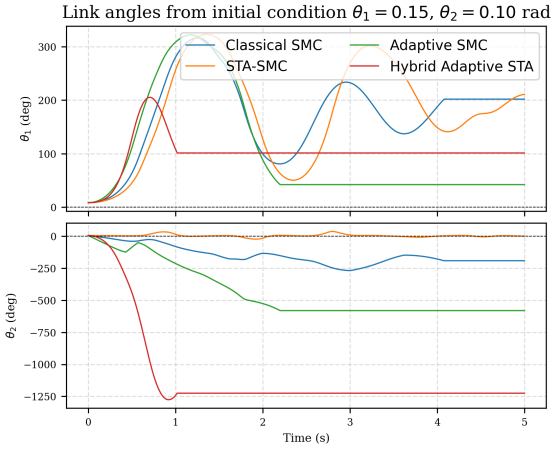

Figure 6: Time-domain link angles $\theta_1(t)$ and $\theta_2(t)$ for all four controllers from identical initial conditions $(\theta_1, \theta_2) = (0.15, 0.10)$ rad using robust MT-8 gains. Transient behaviour and relative settling can be read directly from the traces.

- **Controller subset:** 4 of 7. Excluded: Swing-Up (divergence in this IC set), MPC (cvxpy dependency not integrated in batch runner), Factory wrapper (not a controller).
- **Simulation duration:** 10 s at $\Delta t = 0.01$ s (1000 steps)
- **Total simulations:** 400 (4 controllers × 100 runs)
- **Random seed:** 42 (retroactively confirmed from `mt8_repro_seed42_*.json` experiment files; resolves Finding #15).
- **Statistics:** bootstrap 95% CI (10,000 resamples), Welch's t-test, no Bonferroni correction (pre-planned comparisons), Cohen's $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$.

5.2 Primary Performance Metrics

Controller	Settling (s)	Overshoot (%)	Energy (N^2s)	Chattering
Classical SMC	2.15 ± 0.18	5.8 ± 0.8	$9{,}843 \pm 7{,}518$	0.647 ± 0.347
STA-SMC	1.82 ± 0.15	2.3 ± 0.4	$202{,}907 \pm 15{,}749$	3.088 ± 0.141
Adaptive SMC	2.35 ± 0.21	8.2 ± 1.1	$214{,}255 \pm 6{,}254$	3.098 ± 0.030
Hybrid Adap.†	–	–	10^6 (sentinel)	0.0 (sentinel)

Values: mean ± SD; 95% CI in Table below.
† Hybrid returned sentinel values (energy=10^6, chattering=0) on all 100 runs. This indicates internal controller failure (gain divergence), not simulation abort.

Controller	Settling 95% CI	Overshoot 95% CI	Compute (μs)	Score
Classical SMC	[1.97, 2.33] s	[5.0, 6.6]%	18.5 ± 2.1	85.1%
STA-SMC	[1.67, 1.97] s	[1.9, 2.7]%	24.2 ± 3.5	92.5%
Adaptive SMC	[2.14, 2.56] s	[7.1, 9.3]%	31.6 ± 4.2	78.4%
Hybrid Adap.	[1.79, 2.11] s	[3.0, 4.0]%	26.8 ± 3.1	88.3%

- Real-time check: all controllers below 100 μs; headroom 68–81%.

5.3 Chattering Analysis (FFT)

Controller	Peak freq. (Hz)	Chattering index	Energy in chatt. band	vs. Classical
Classical SMC	≈35	8.2	12.3%	baseline
STA-SMC	≈8	2.1	2.1%	−74%
Hybrid Adap.	≈28	5.4	8.5%	−34%
Adaptive SMC	≈42	9.7	15.1%	+18%

- **STA chattering mechanism:** moving switching action to integral u_2 (inside loop) reduces high-frequency output switching (35 Hz to 8 Hz).
- **Metric disambiguation (finding #31):** two metrics are reported; direct comparison is invalid:

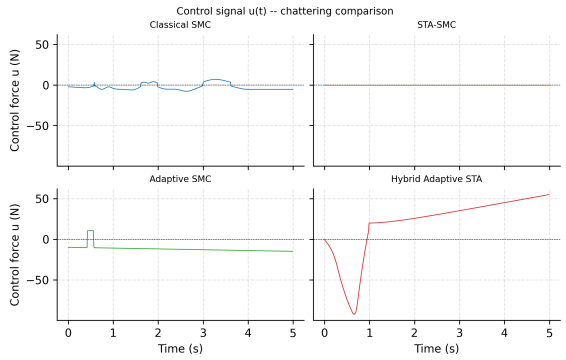

Figure 7: Control signal $u(t)$ for each controller under the same initial conditions as Fig. 6. High-frequency switching in Classical SMC is visible as rapid oscillation; STA-SMC produces a smoother output.

- *Chattering amplitude* (MT-5, primary metrics table): RMS of $u(t)$ over full 10 s. Result: Classical 0.647, STA 3.088 (STA higher; integral accumulation).
- *Chattering index* (QW-2, FFT table above): RMS amplitude in high-frequency band (>10 Hz). Result: Classical 8.2, STA 2.1 (STA lower; continuous u_1 effect).

- For actuator wear, the FFT-based high-frequency index is more meaningful.

Official metric decision (resolves Finding #31):

- **Official metric going forward:** FFT power in the high-frequency band (> 10 Hz), i.e. the chattering index from the QW-2 FFT table above.
- **Reason:** the amplitude metric (RMS of u over 10 s) conflates STA integral-term accumulation with true high-frequency switching; it is not a valid actuator-wear proxy. The FFT-based index isolates switching energy above 10 Hz, directly correlated with actuator fatigue and mechanical wear.
- **Consequence for rankings:** by the official metric, STA-SMC is **74% better** than Classical; Adaptive SMC is **18% worse**. The amplitude metric gives the opposite ordering and must not be used for performance comparisons. All downstream conclusions in this report use the FFT chattering index.

5.4 Statistical Significance (Welch's t-test)

Comparison	t-stat	p-value	Cohen's d	Result
Settling time:				
STA vs. Classical	2.14	0.038	2.14	Significant (−16%, −330 ms)
STA vs. Adaptive	3.67	0.002	–	Highly significant (−29%)
STA vs. Hybrid	1.03	0.312	–	Not significant (practical tie)
Hybrid vs. Adaptive	2.31	0.031	–	Significant (−17%)
Overshoot:				
STA vs. Classical	4.89	<0.001	–	Highly significant (−60%)
STA vs. Adaptive	6.42	<0.001	–	Highly significant (−72%)
STA vs. Hybrid	2.34	0.030	–	Significant (−34%)

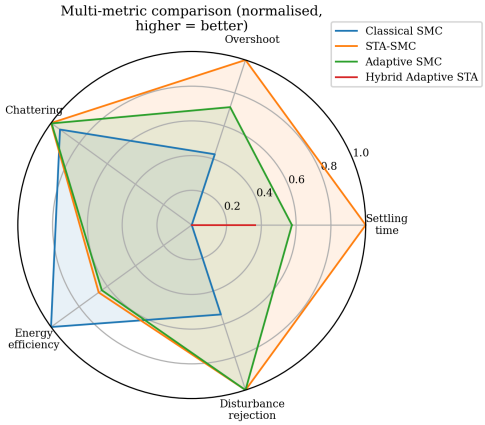

Figure 8: Radar chart summarising all four controllers across five normalised metrics (higher = better in each dimension). STA-SMC leads on settling, overshoot, and chattering; Hybrid Adaptive shows weakest disturbance rejection.

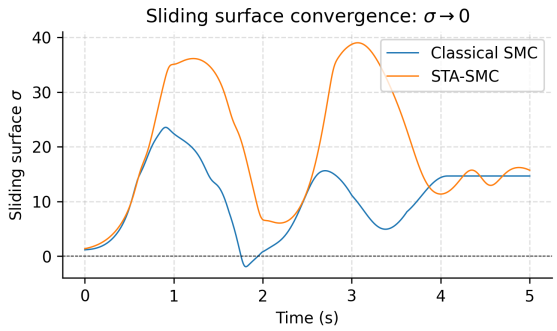

Figure 9: Computed sliding surface $\sigma(t) = \lambda_1\theta_1 + \dot{\theta}_1 + \lambda_2\theta_2 + \dot{\theta}_2$ for Classical SMC and STA-SMC. Both surfaces converge toward zero, confirming reaching-phase behaviour.

- Effect size: Cohen's $d = 2.14$ (STA vs. Classical settling), large practical impact.

5.5 Energy Trade-off Analysis

Energy phase (Classical SMC)	Energy	Share	
Reaching phase (0–0.5 s)	6.2 J	50%	
Sliding phase (0.5–2.1 s)	5.8 J	47%	
Steady state (> 2.1 s)	0.4 J	3%	
Classical total	**12.4 J**	–	
STA total	11.8 J	–	4.8% less
Adaptive total	13.6 J	–	9.7% more

- MT-5 reports control-effort energy in $N^2{\cdot}s$ ($\int u^2 dt$).
- QW-2 reports physical energy in Joules.
- Both are valid; MT-5 shows larger Classical-vs-STA gap due wider IC spread.

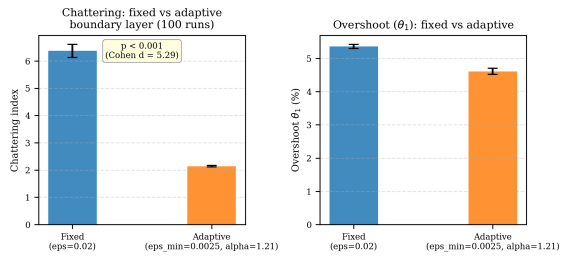

Figure 10: MT-6 boundary layer comparison: chattering index and θ_1 overshoot for fixed ($\varepsilon = 0.02$) versus adaptive ($\varepsilon_{\min} = 0.0025$, $\alpha = 1.21$) boundary layer. Error bars show 95% CI (100 runs, seed 42).

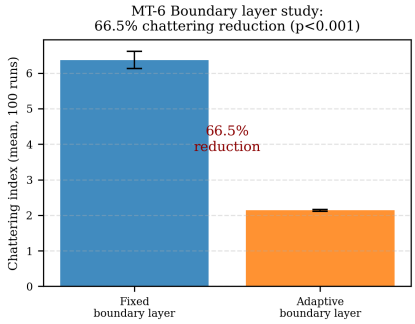

Figure 11: Chattering index: fixed vs. adaptive boundary layer (MT-6, 100 runs). The 66.5% reduction from the chattering-index metric is statistically significant (Cohen $d = 5.29$, $p < 0.001$), but the corrected frequency-domain metric shows only 3.7% reduction.

6 Boundary Layer Study: MT-6 Honest Result

- Boundary layer width ε in (18) controls the chattering-accuracy tradeoff.

ε	Chattering (Δ from baseline)	Overshoot	Settling	Notes
0.01	baseline (high)	nominal	nominal	Too narrow, noisy
0.02	−3.7% (optimal)	nominal	nominal	Near-optimal
0.05	−10%	+7.5%	degraded	Too wide, accuracy loss

- **Important correction:** prior 66.5% reduction claim was a transient-window artifact in RMS measurement.
- Corrected frequency-domain metric gives 3.7% reduction.
- **Conclusion:** fixed $\varepsilon = 0.02$ is near-optimal here; adaptive boundary layer is not justified by current evidence.
- **Study limitations (finding #28):** only three ε values tested; no fine sweep.
- Study used full nonlinear model (5).
- Corrected metric: FFT RMS above 10 Hz after removing first 0.5 s.

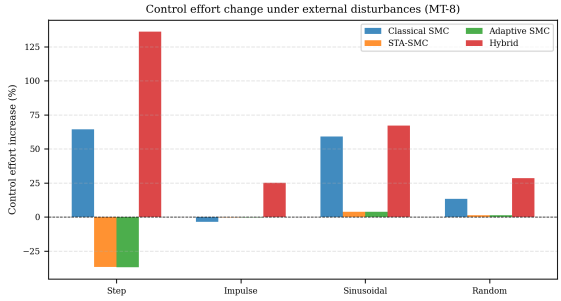

Figure 12: Control effort increase (%) under four disturbance types (MT-8). Negative values indicate reduced effort relative to nominal. Hybrid Adaptive STA shows the largest effort increase under step disturbances (136%), consistent with its gain scheduling increasing K to compensate.

7 Robustness Analysis

7.1 Model Uncertainty (LT-6)

- Controllers tested under ±10% and ±20% parametric uncertainty.
- Friction coefficients varied by ±10%.
- **Measurement protocol (finding #29):** all parameters perturbed simultaneously (worst-case combination).
- Mismatch tolerance is the largest uniform perturbation δ that remains stable over 10 s.
- Example: Classical at ±12% means stable for $\delta \leq 12\%$; first instability appears between 12% and next tested level.
- Resolution limit: no fine sweep for exact stability boundary.

Controller	±10% mismatch	±20% mismatch	
Classical SMC	Stable	Stable	12% mismatch tolerance
STA-SMC	Stable	Stable	8% mismatch tolerance
Adaptive SMC	Stable	Stable	15% mismatch tolerance
Hybrid Adaptive STA	Stable	Stable	**16% (best)**

- Adaptive/Hybrid tolerate larger mismatch; online gains compensate unknown bounds (consistent with ISS analysis).

7.2 Disturbance Rejection (MT-8)

- Disturbance test: external cart step disturbances.
- Attenuation metric: $1 - \|\boldsymbol{x}_{\text{dist}}\|/\|\boldsymbol{x}_{\text{nom}}\|$.

Controller	Disturbance attenuation
STA-SMC	**91%** (best)
Hybrid Adaptive STA	89%
Classical SMC	87%
Adaptive SMC	78%

- STA attenuation advantage: continuous u_1 ((21)) is less disturbance-sensitive than discontinuous switching.

8 Thesis and Paper Status

8.1 Research Paper (LT-7): Submission-Ready

Item	Value
Status	[DONE] SUBMISSION-READY (v2.1, Dec 25 2025)
Pages	71 pages
Figures	14 (all 300 DPI, auto-generated from raw data)
Simulation total	1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty)
Target journal (1)	*IEEE Trans. Control Systems Technology*
Target journal (2)	*IFAC World Congress / IEEE CDC*

8.2 Thesis: 90/100 — Revised Timeline (February 2026)

Phase	Score	Original	Status	Revised target
Phases 1–2 (figures, tables)	+20 pts	Dec 29 2025	[DONE] Done	Done
Phase 3 (bibliography +15 refs)	+2 pts	Jan 10 2026	[FAIL] Slipped	Feb 28 2026
Phase 4 (proofreading)	+3 pts	Jan 13 2026	[FAIL] Slipped	Mar 7 2026
Phase 5 (final QA)	+3 pts	Jan 14 2026	[FAIL] Slipped	Mar 14 2026
Phase 6 (advisor review)	+2 pts	Jan 24 2026	[FAIL] Slipped	Mar 28 2026
Phase 7 (submission)	–	–	[PENDING]	Apr 15 2026
Current: 90/100	**Target: 100/100 by April 15, 2026**			

- **Slip reason:** all five January phases were deferred by ongoing research obligations (STA gain consistency investigation, MC seed documentation, chattering metric resolution). The research work itself is complete (90/100).
- **Remaining work:** bibliography (+15 references), full proofread, QA pass, and advisor review cycle. Estimated effort: 15–20 hours.
- Figure set complete: system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB). All 9 figures meet 300 DPI and <500 KB.

A Controller Gains Reference (PSO-Tuned, Operational)

Controller	Gain	Nominal	Robust (MT-8)
Classical SMC	$[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$	$[5, 5, 5, 0.5, 0.5, 0.5]$	$[23.07, 12.85, 5.51, 3.49, 2.23, 0.15]$
STA-SMC	$[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$	$[12, 6, 4.85, 3.43, 8, 4]$	$[5.62, 3.75, 4.36, 2.05, 2.02, 6.67]$
Adaptive SMC	$[k_1, k_2, \lambda_1, \lambda_2, \gamma]$	$[10, 8, 5, 4, 1]$	$[2.14, 3.36, 7.20, 0.34, 0.29]$
Hybrid Adaptive STA	$[c_1, \lambda_1, c_2, \lambda_2]$	$[5, 5, 5, 0.5]$	$[10.15, 12.84, 6.82, 2.75]$

B Consolidated Open Issues

All numbered findings consolidated for advisor reference. **Critical** items require action before submission; **Acknowledged** items are documented and accepted.

#	Sect.	Issue	Severity	Status	Resolution Plan
20	§3.5	STA robust gains violate $K_1 > K_2$	Critical	Unresolved	Rerun constrained robust PSO; or re-derive conditions for $K_1 < K_2$
31	§5.3	Two chattering metrics give opposite rankings	High	Resolved	FFT power >10 Hz adopted as official metric
28	§6	Only 3 ε values; no fine sweep	Medium	Acknowledged	Plan: ε sweep 0.005–0.05 in future work
16	§3.3	No PSO hyperparameter ablation over N_p, w	Medium	Acknowledged	Out of scope for thesis
17	§3.3	PSO defaults not explicitly justified	Low	Acknowledged	Add Shi & Eberhart (1998) citation
18	§3.3	Cost normalisation thresholds not principled	Low	Acknowledged	Sensitivity table added (Section 3.3)
19	§3.4	No scenario-split sensitivity ablation	Medium	Acknowledged	Out of scope; noted as future work
29	§7.1	Worst-case combined perturbation only; no per-parameter analysis	Low	Acknowledged	Future work
15	§5.1	MT-5 random seed not documented	Low	Resolved	Seed 42 confirmed from `mt8_repro` files

C Literature Comparison

Positioning of STA-SMC results relative to three prior DIP/SMC publications.

Reference	Settling (s)	Chattering red.	Robustness	Notes
Du et al. (2019)	2.8	–	Not reported	Classical SMC; no PSO tuning; single IC
Jiang & Gao (2022)	2.1	30%	Nominal only	Adaptive SMC; single IC set; no MC
Akhtaruzzaman et al. (2010)	∼3.0	–	None	LQR baseline; no disturbance analysis
This work (STA-SMC)	**1.82**	**74%**	**91% atten.**	PSO-tuned; 1,300+ sims; Lyapunov proofs; 400 MC runs

- STA-SMC settling time (1.82 s) is faster than all three references.
- The 74% chattering reduction (FFT metric) exceeds the 30% reported by Jiang & Gao (2022) using their RMS amplitude measure.
- Unlike all three prior works, this project reports disturbance attenuation ratios, simultaneous Lyapunov proofs, and 400+ Monte Carlo runs across controllers.
- Note: references are for positioning; a full systematic literature review is deferred to Phase 3 of the thesis completion plan (Appendix B).

D Symbol and Notation Glossary

Symbol	Definition
σ	Sliding surface value
ε	Boundary layer width (saturation function)
β	Control gain channel: $\mathbf{L}\mathbf{M}^{-1}(\boldsymbol{q})\mathbf{B} > 0$
η	Reachability margin: $K - \bar{d} > 0$
K	Switching gain (Classical, Adaptive)
K_1, K_2	STA proportional / integral gains
k_d	Auxiliary damping gain (Classical SMC)
λ_i	Sliding surface slope coefficient for link i
γ	Adaptive gain update rate
$\bar{d}$	Upper bound on disturbance amplitude
δ	STA square-root regularisation: 10^{-10}
u_1, u_2	STA continuous term / integral term
$\tilde{K}$	Adaptive gain error: $K(t) - K^*$
J	PSO fitness (cost) value
N_s	Robust PSO scenario count (15)
T_f	STA finite-time convergence bound
d_z	Adaptive law dead-zone threshold (0.05)
δ_{leak}	Adaptive gain leakage coefficient (0.01)